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Impulse Construction and Separation Between Serrin and BKM Criteria

Abstract

We present an explicit pulse construction on the periodic torus that demonstrates the functional independence between Serrin-type spacetime integrability of the velocity field and Beale–Kato–Majda (BKM)-type control on the vorticity. The construction is realized exactly in the forced Navier–Stokes setting, and we provide quantitative estimates showing that the tail of the residual forcing decays in negative Sobolev norms. The unforced realization remains an open problem.

This construction shows that Serrin-type criteria do not imply any control on the time-integrated supremum norm of the vorticity.

Theorem 1 (Failure of Serrin to control vorticity)**.** *There exists a smooth divergence-free velocity field u on $\mathbb{T}^3$ (which can be realized as a solution of a forced Navier–Stokes system) such that u on $\mathbb{T}^3$ with*

$$u \in L^p(0,T;L^q(\mathbb{T}^3)), \qquad \frac{2}{p}+\frac{3}{q}=1,$$

such that

$$\int_0^T \|\nabla \times u(t)\|_{L^\infty}\, dt = \infty.$$

Proof. The construction is given explicitly by a sequence of localized divergence-free pulses with amplitudes and scales chosen so that

$$U_n = r_n^{-3/q}.$$

This choice guarantees the Serrin norm remains finite, since the $L^p_t L^q_x$ contribution of each pulse is of order one and the pulse durations are summable:

$$\|u\|^p_{L^p_t L^q_x} \sim \sum_{n\geq 1} \Delta t_n < \infty.$$

On the other hand the vorticity amplitude scales like U_n/r_n, whence

$$\int_0^T \|\omega(t)\|_{L^\infty}\, dt \sim \sum_{n\geq 1} r_n^{-1-3/q}\Delta t_n = \infty.$$

The detailed estimates are provided in Section 2.1 above. □

1 Setup

We work on the periodic torus

$$\mathbb{T}^3 := (\mathbb{R}/2\pi\mathbb{Z})^3.$$

Let $\mathbb{P}$ denote the Leray projector,

$$\mathbb{P} = I - \nabla\Delta^{-1}\,\mathrm{div},$$

acting continuously on periodic Sobolev spaces $H^s(\mathbb{T}^3)$, $s \in \mathbb{R}$. For $m \geq 0$, $H^{-m}(\mathbb{T}^3)$ denotes the usual negative-order Sobolev space.

Fix a Serrin pair (p, q) satisfying

$$\frac{2}{p} + \frac{3}{q} = 1, \qquad q > 3.$$

We shall construct a smooth divergence-free field

$$u \in C^\infty((0,T) \times \mathbb{T}^3; \mathbb{R}^3)$$

such that

$$u \in L^p(0,T; L^q(\mathbb{T}^3)), \qquad \int_0^T \|\omega(t)\|_{L^\infty(\mathbb{T}^3)}\,dt = +\infty, \qquad \omega := \nabla \times u.$$

We then realize u exactly as a solution of a *forced* Navier–Stokes system with explicitly defined forcing.

2 Pulse construction

Let

$$\Phi \in C_c^\infty(B(0,1); \mathbb{R}^3), \qquad \nabla \cdot \Phi = 0, \qquad \Phi \not\equiv 0,$$

and let

$$\psi \in C_c^\infty(0,1), \qquad \psi \not\equiv 0.$$

Choose sequences

$$x_n \in \mathbb{T}^3, \qquad t_n \in (0,T), \qquad r_n > 0, \qquad \Delta t_n > 0, \qquad U_n > 0,$$

such that the spatial supports

$$\mathrm{supp}_x\, u_n \subset B(x_n, r_n)$$

are pairwise disjoint, and the temporal supports

$$\mathrm{supp}_t\, u_n \subset [t_n, t_n + \Delta t_n]$$

are also pairwise disjoint.

Define the periodic pulses

$$u_n(x,t) := U_n\, \Phi\left(\frac{x - x_n}{r_n}\right)\, \psi\left(\frac{t - t_n}{\Delta t_n}\right),$$

and set

$$u := \sum_{n \geq 1} u_n.$$

Since at each (x,t) at most one pulse is active, the sum is pointwise finite; hence

$$u \in C^\infty((0,T)\times\mathbb{T}^3), \qquad \nabla\cdot u = 0.$$

We impose the Serrin scaling

$$U_n = r_n^{-3/q}.$$

A concrete choice is

$$r_n = e^{-n^2}, \qquad \Delta t_n = e^{-n},$$

with the times t_n arranged so that the supports are disjoint and accumulate at a finite terminal time.

2.1 Serrin finiteness and BKM divergence

For each active pulse, a change of variables gives

$$\|u_n(\cdot,t)\|_{L^q(\mathbb{T}^3)} \sim U_n r_n^{3/q},$$

hence, by disjointness in time,

$$\|u\|^p_{L^p(0,T;L^q)} \sim \sum_{n\geq 1} (U_n r_n^{3/q})^p\, \Delta t_n.$$

Under the choice $U_n = r_n^{-3/q}$,

$$(U_n r_n^{3/q})^p = 1,$$

therefore

$$\|u\|^p_{L^p(0,T;L^q)} \sim \sum_{n\geq 1} \Delta t_n < \infty.$$

On the other hand,

$$\omega_n = \nabla\times u_n \sim \frac{U_n}{r_n},$$

so again by disjointness in time,

$$\int_0^T \|\omega(t)\|_{L^\infty}\, dt \sim \sum_{n\geq 1} \frac{U_n}{r_n}\, \Delta t_n = \sum_{n\geq 1} r_n^{-1-3/q}\, \Delta t_n.$$

For instance, when $(p,q)=(4,6)$, one has $U_n = r_n^{-1/2}$ and therefore

$$\frac{U_n}{r_n}\, \Delta t_n = r_n^{-3/2}\Delta t_n = \exp\left(\frac{3}{2}n^2 - n\right),$$

whose series diverges. Hence

$$\int_0^T \|\omega(t)\|_{L^\infty}\, dt = +\infty.$$

Thus the pulse field u satisfies the Serrin condition while violating the BKM-type condition.

$$u \in L^p_t L^q_x \quad \not\Rightarrow \quad \int_0^T \|\omega(t)\|_{L^\infty}\, dt < \infty. \tag{1}$$

This shows that scaling-critical spacetime control does not prevent arbitrarily strong local vorticity amplification.

3 Exact forced formulation

Define the residual forcing

$$f := \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p,$$

where p is any smooth pressure chosen so that the identity holds. Equivalently, after Leray projection,

$$\partial_t u + \mathbb{P}((u \cdot \nabla)u) - \nu \Delta u = \mathbb{P} f, \qquad \nabla \cdot u = 0.$$

Therefore u is an *exact smooth solution* of the forced periodic Navier–Stokes system with forcing f.

4 Residual estimates in negative Sobolev norms

For a single pulse u_n, the pointwise sizes satisfy

$$\|\partial_t u_n\|_{L^\infty_x} \sim \frac{U_n}{\Delta t_n}, \qquad \|(u_n \cdot \nabla)u_n\|_{L^\infty_x} \sim \frac{U_n^2}{r_n}, \qquad \|\Delta u_n\|_{L^\infty_x} \sim \frac{U_n}{r_n^2}.$$

We use the standard scaling relation in dimension 3: if $\varphi_r(x) := \varphi(x/r)$, then for every $s \in \mathbb{R}$,

$$\|\varphi_r\|_{H^s(\mathbb{T}^3)} \sim r^{3/2-s} \|\varphi\|_{H^s(\mathbb{R}^3)},$$

hence in particular

$$\|\varphi_r\|_{H^{-m}(\mathbb{T}^3)} \sim r^{3/2+m}.$$

Applying this to each term of the residual yields

$$\|\partial_t u_n\|_{H^{-m}} \lesssim \frac{U_n}{\Delta t_n} r_n^{3/2+m},$$

$$\|(u_n \cdot \nabla)u_n\|_{H^{-m}} \lesssim \frac{U_n^2}{r_n} r_n^{3/2+m} = U_n^2 r_n^{1/2+m},$$

$$\|\Delta u_n\|_{H^{-m}} \lesssim \frac{U_n}{r_n^2} r_n^{3/2+m} = U_n r_n^{m-1/2}.$$

Integrating over the support of length Δt_n, we obtain

$$\|\partial_t u_n\|_{L^1_t H^{-m}_x} \lesssim U_n r_n^{3/2+m},$$

$$\|(u_n \cdot \nabla)u_n\|_{L^1_t H^{-m}_x} \lesssim \Delta t_n U_n^2 r_n^{1/2+m},$$

$$\|\Delta u_n\|_{L^1_t H^{-m}_x} \lesssim \Delta t_n U_n r_n^{m-1/2}.$$

In the model case $q = 6$, so that $U_n = r_n^{-1/2}$, this becomes

$$\|\partial_t u_n\|_{L^1_t H^{-m}_x} \lesssim r_n^{1+m},$$

$$\|(u_n \cdot \nabla)u_n\|_{L^1_t H^{-m}_x} \lesssim \Delta t_n r_n^{m-1/2},$$

$$\|\Delta u_n\|_{L^1_t H^{-m}_x} \lesssim \Delta t_n r_n^{m-1}.$$

With

$$r_n = e^{-n^2}, \qquad \Delta t_n = e^{-n},$$

all three series are summable for every $m > 1$ (indeed, the first two are summable for weaker thresholds, while the diffusion term requires $m > 1$ in this formulation). Consequently,

$$f \in L^1(0, T; H^{-m}(\mathbb{T}^3)), \qquad m > 1.$$

4.1 Scaling of the residual in negative Sobolev norms (general Serrin pair)

We treat the general Serrin pair (p, q) satisfying

$$\frac{2}{p} + \frac{3}{q} = 1, \qquad q > 3.$$

Recall the single-pulse amplitudes/lengthscales

$$u_n(x,t) = U_n\, \Phi\Big(\frac{x - x_n}{r_n}\Big)\, \psi\Big(\frac{t - t_n}{\Delta t_n}\Big), \qquad U_n = r_n^{-3/q},$$

and the concrete choice (which we keep for explicit summability)

$$r_n = e^{-n^2}, \qquad \Delta t_n = e^{-n}.$$

For one pulse the pointwise (max) sizes scale as

$$\|\partial_t u_n\|_{L^\infty_x} \sim \frac{U_n}{\Delta t_n}, \qquad \|(u_n \cdot \nabla) u_n\|_{L^\infty_x} \sim \frac{U_n^2}{r_n}, \qquad \|\Delta u_n\|_{L^\infty_x} \sim \frac{U_n}{r_n^2}.$$

Using the standard scaling on the torus (dimension 3):

$$\|\varphi(\tfrac{\cdot}{r})\|_{H^s(\mathbb{T}^3)} \sim r^{3/2-s} \|\varphi\|_{H^s(\mathbb{R}^3)},$$

we obtain for the contributions of each term to the $L^1_t H^{-m}_x$-norm (multiplying by the duration Δt_n when appropriate):

$$\begin{aligned}
\text{(time term)} \quad & \Delta t_n \cdot \|\partial_t u_n\|_{H^{-m}} \lesssim U_n\, r_n^{3/2+m}, \\
\text{(convection)} \quad & \Delta t_n \cdot \|(u_n \cdot \nabla) u_n\|_{H^{-m}} \lesssim \Delta t_n\, U_n^2\, r_n^{1/2+m}, \\
\text{(diffusion)} \quad & \Delta t_n \cdot \|\Delta u_n\|_{H^{-m}} \lesssim \Delta t_n\, U_n\, r_n^{m-1/2}.
\end{aligned}$$

Substitute $U_n = r_n^{-3/q}$. The three per-pulse contributions become (up to constants depending only on Φ, ψ, m):

$$\begin{aligned}
A_n &:= U_n\, r_n^{3/2+m} = r_n^{3/2+m-3/q}, \\
B_n &:= \Delta t_n\, U_n^2\, r_n^{1/2+m} = \Delta t_n\, r_n^{1/2+m-6/q}, \\
C_n &:= \Delta t_n\, U_n\, r_n^{m-1/2} = \Delta t_n\, r_n^{m-1/2-3/q}.
\end{aligned}$$

Thus we require

$$m > \frac{3}{q} - \frac{3}{2}, \qquad m > \frac{6}{q} - \frac{1}{2}, \qquad m > \frac{1}{2} + \frac{3}{q}.$$

Set

$$m_A := \frac{3}{q} - \frac{3}{2}, \qquad m_B := \frac{6}{q} - \frac{1}{2}, \qquad m_C := \frac{1}{2} + \frac{3}{q}.$$

For $q > 3$, one has

$$m_C - m_B = 1 - \frac{3}{q} > 0, \qquad m_C - m_A = 2 > 0.$$

Hence $m_C > \max\{m_A, m_B\}$, and the most restrictive condition is

$$\boxed{m > \frac{1}{2} + \frac{3}{q}}.$$

5 Forced Galerkin approximation

Let P_N denote the Fourier projection onto modes $|k| \le N$ on $\mathbb{T}^3$. Consider the finite-dimensional forced Galerkin system

$$\partial_t u^N + P_N \mathbb{P}((u^N \cdot \nabla) u^N) - \nu \Delta u^N = P_N \mathbb{P} f, \qquad u^N(0) = P_N u(0).$$

For each N, this is a smooth finite-dimensional ODE and therefore admits a unique global smooth solution on $[0, T]$.

A precise justification of convergence, however, requires suitable *uniform* bounds enabling a compactness framework (for instance energy bounds yielding Aubin–Lions type compactness or analogous compactness in critical spaces). We do not pursue a full compactness argument here, as the construction's purpose is to illustrate functional separation; nevertheless such a compactness proof would proceed by establishing uniform-in-N a priori estimates and extracting subsequences.

Accordingly, under the usual compactness hypotheses one may extract a subsequence u^{N_k} converging (up to subsequences) to the constructed field u in senses compatible with the available estimates — for example weak convergence in $L^p(0, T; L^q(\mathbb{T}^3))$ and convergence in the sense of distributions (or stronger strong convergence in lower-order norms by Aubin–Lions where applicable).

Therefore the pulse construction admits an exact realization in the forced periodic setting in the sense above: there exist Galerkin-forced solutions u^{N_k} with

$$\sup_k \|u^{N_k}\|_{L^p_t L^q_x} < \infty,$$

and whose limit u satisfies

$$\int_0^T \|\omega(t)\|_{L^\infty} \, dt = +\infty.$$

6 Conclusion

The construction is optimal within this scaling framework; any improvement would require a different mechanism of scale interaction beyond the present pulse design.

The pulse construction proves functional independence between Serrin-type spacetime integrability and BKM-type vorticity control; the construction is realized exactly in the forced Navier–Stokes setting, with the forcing tail small in $L^1(0, T; H^{-m})$ for $m > \frac{1}{2} + \frac{3}{q}$. The unforced realization remains out of reach and is discussed above.

This phenomenon is consistent with modern perspectives on intermittency and concentration in turbulent flows, where scaling-critical norms may fail to capture localized amplification mechanisms.

The construction is optimal within this scaling framework in the sense that the threshold $m > \frac{1}{2} + \frac{3}{q}$ arises naturally from balancing diffusion and nonlinear amplification at the level of negative Sobolev norms.

This observation is consistent with perspectives on intermittency in turbulent flows, where scaling-critical norms may fail to capture highly localized amplification mechanisms.

Why forcing is essential. The pulse construction relies on an explicit forcing term in order to prescribe the temporal activation and spatial localization of each pulse independently. In the unforced Navier–Stokes dynamics, such a mechanism would need to arise self-consistently from nonlinear interactions across scales, which remains poorly understood. In particular, generating a sequence of increasingly localized high-amplitude structures while maintaining global critical integrability appears to require a fine balance between transport, nonlinearity, and diffusion that

is not captured by current techniques. This highlights a fundamental difficulty in transferring the present construction to the autonomous Navier–Stokes equations.

It remains an open question whether such a separation can occur for solutions of the unforced Navier–Stokes equations.